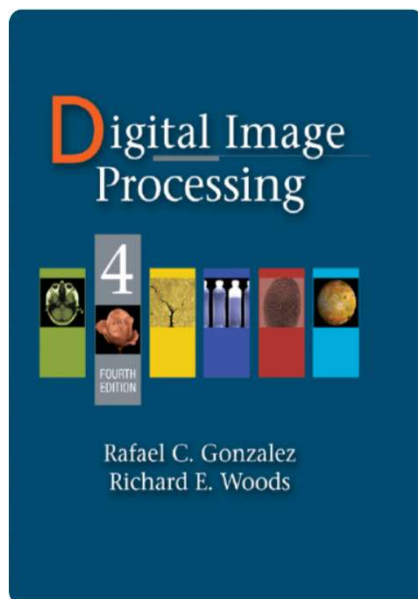




1



Source:

Chapter 3: Intensity Transformations and Spatial Filtering

- Title: Digital Image Processing
- Authors:
 - Rafael C. Gonzalez, University of Tennessee,
 - Richard E. Woods, Interapptics.
- Publisher: New York, NY : Pearson, [2018]

2

Objectives

- Understand the meaning of spatial domain processing, and how it differs from transform domain processing.
- Be familiar with the principal techniques used for intensity transformations.
- Understand the physical meaning of image histograms and how they can be manipulated for image enhancement.
- Understand the mechanics of spatial filtering, and how spatial filters are formed.
- Understand the principles of spatial convolution and correlation.
- Be familiar with the principal types of spatial filters, and how they are applied.
- Be aware of the relationships between spatial filters, and the fundamental role of lowpass filters.
- Understand the principles of fuzzy logic and how these principles apply to digital image processing.

3

Enhancement

- Goal: to modify an image so that its utilization on a particular application is enhanced.
- A set of ad hoc tools applicable based on viewer's specific needs.
- No general theory on image enhancement exists.
- Methods:
 - Spatial domain
 - Pixel processing
 - Gray level transformation: Data independent
 - histogram processing: Data-dependent
 - Arithmetic ops
 - Spatial filtering
 - Frequency domain filtering

4

Basics of Intensity Transformations

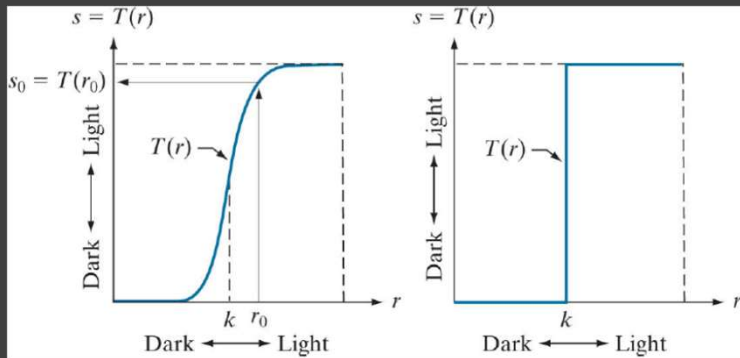


FIGURE 3.2: Intensity transformation functions. (a) Contrast stretching function. (b) Thresholding function

5

3.2 Some Basic Intensity Transformation Functions

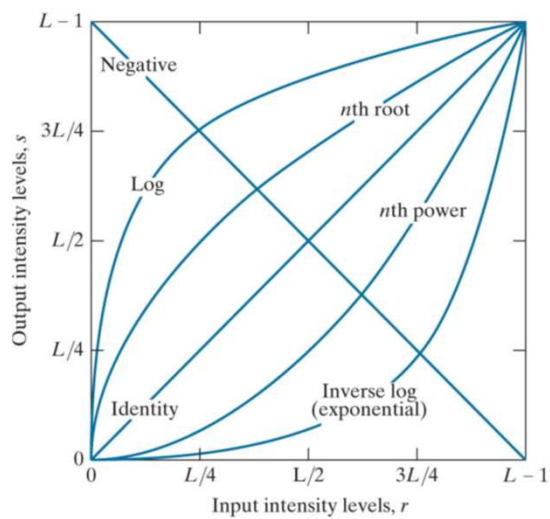
- Operated on individual pixel's intensity values: $s = T(r)$. r : original intensity, s : new intensity
- Data independent pixel-based enhancement method.

Methods

- Image Negative
- Log Transformation
- Power-Law (Gamma) Transformations
- Piecewise Linear Transformation
- Functions Intensity Slicing
- Bit Slicing



6

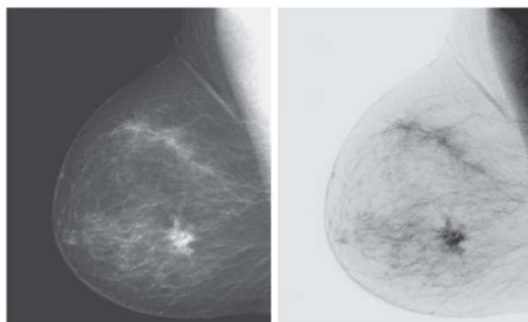


Basic intensity transformation functions

FIGURE 3.3

Some basic intensity transformation functions. Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.

7



$$s = L - 1 - r$$

Image Negatives

Suitable for enhancing white or gray details in dark background

FIGURE 3.4

(a) A digital mammogram. (b) Negative image obtained using **Eq. (3-3)**. (Image (a) Courtesy of General Electric Medical Systems.)

8

$$s = c \log(1 + r)$$

Log Transformations

expand dark value to enhance details of dark area

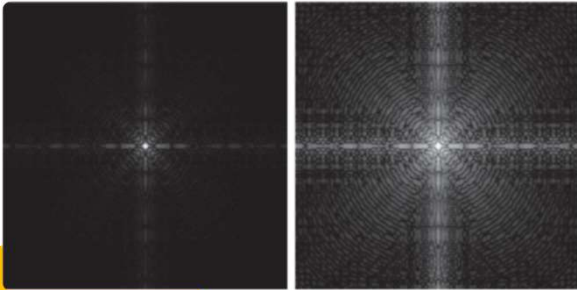
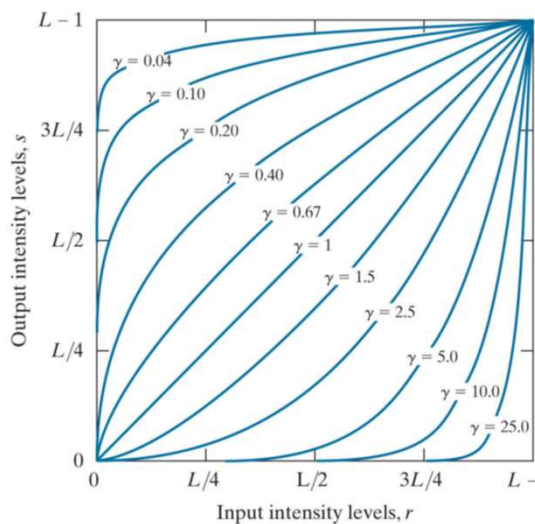


FIGURE 3.5

(a) Fourier spectrum displayed as a grayscale image. (b) Result of applying the log transformation in **Eq. (3-4)** with $c=1$. Both images are scaled to the range $[0, 255]$.

9



$$s = cr^\gamma$$

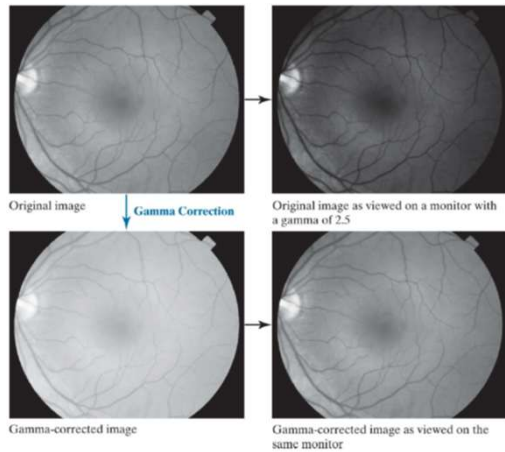
Power-Law (Gamma) Transformations

Gamma correction: to compensate the built-in power law compression due to display characteristics.

FIGURE 3.6

Plots of the gamma equation for various values of γ ($c = 1$ in all cases). Each curve was scaled *independently* so that all curves would fit in the same graph. Our interest here is on the *shapes* of the curves, not on their relative values.

10

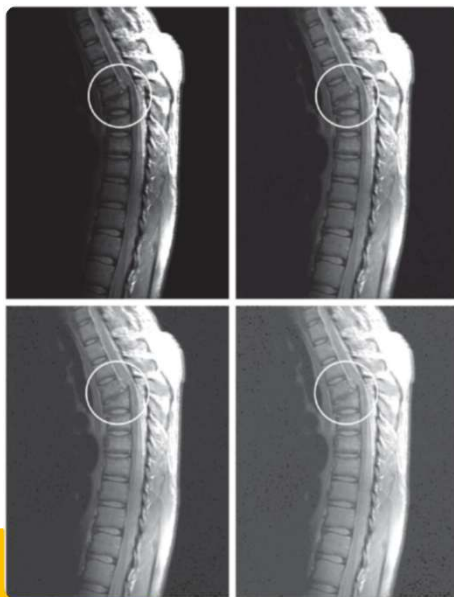


Power-Law (Gamma) Transformations

FIGURE 3.7

(a) Image of a human retina. (b) Image as it appears on a monitor with a gamma setting of 2.5 (note the darkness). (c) Gamma-corrected image. (d) Corrected image, as it appears on the same monitor (compare with the original image). (Image (a) courtesy of the National Eye Institute, NIH)

11

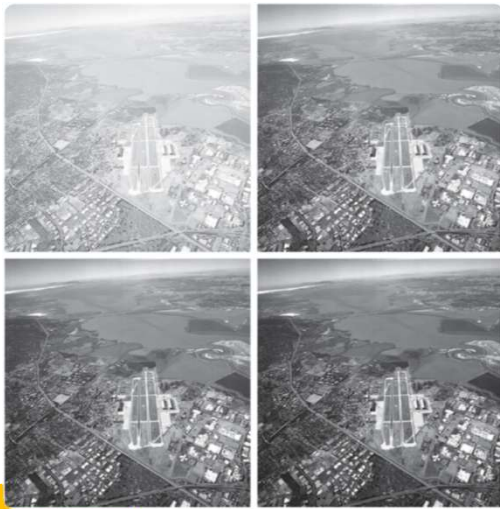


Contrast enhancement using power-law intensity transformations

FIGURE 3.8

(a) Magnetic resonance image (MRI) of a fractured human spine (the region of the fracture is enclosed by the circle). (b)–(d) Results of applying the transformation in **Eq. (3-5)** with $c = 1$ and $\gamma = 0.6, 0.4$, and 0.3 , respectively. (Original image courtesy of Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

12

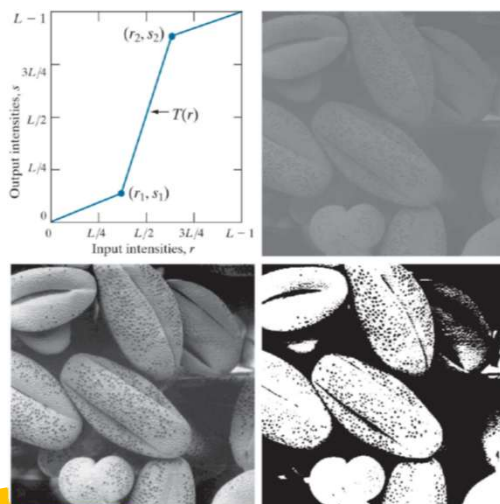


power-law intensity transformations

FIGURE 3.9

(a) Aerial image. (b)–(d) Results of applying the transformation in **Eq. (3-5)** with $\gamma = 3.0, 4.0$, and 5.0 , respectively. ($c = 1$ in all cases.) (Original image courtesy of NASA.)

13

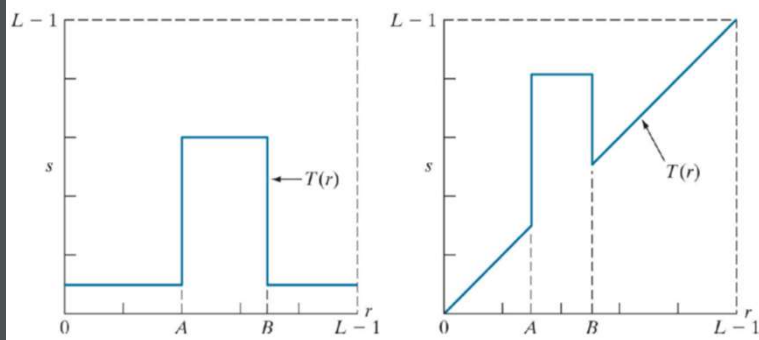


Piecewise Linear Transformation Functions

FIGURE 3.10

Contrast stretching. (a) Piecewise linear transformation function. (b) A low-contrast electron microscope image of pollen, magnified 700 times. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

14



Intensity-Level Slicing

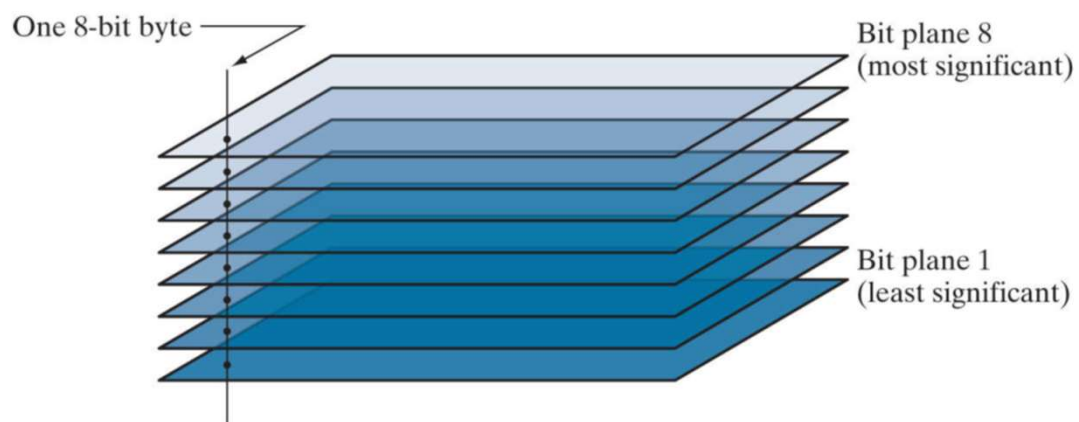
FIGURE 3.11

(a) This transformation function highlights range $[A, B]$ and reduces all other intensities to a lower level. (b) This function highlights range $[A, B]$ and leaves other intensities unchanged.

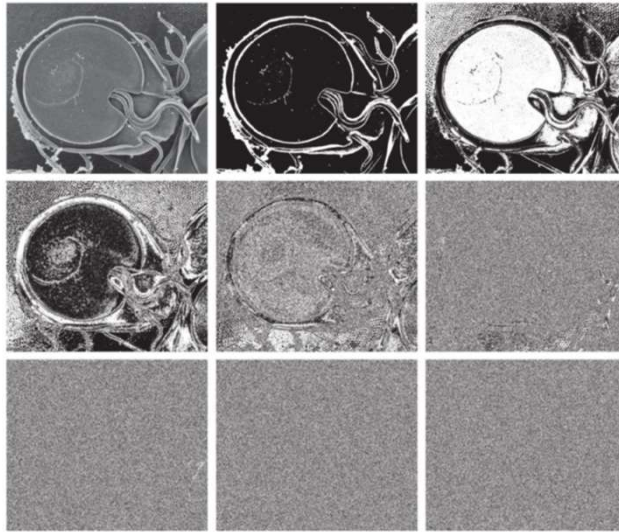
15

Bit-Plane Slicing

FIGURE 3.13: Bit-planes of an 8-bit image.



16

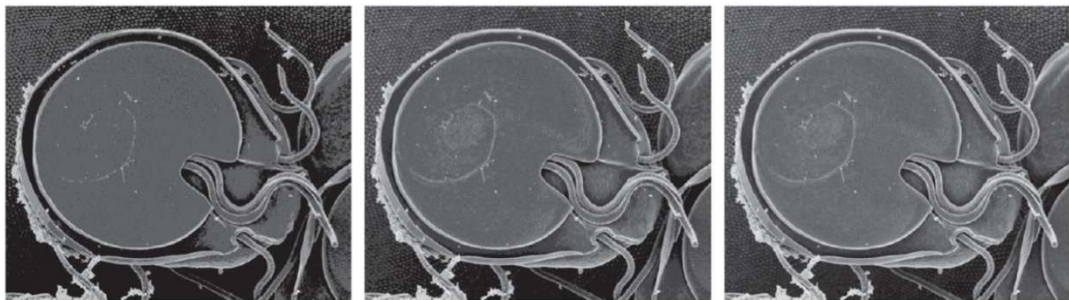


Bit-Plane Slicing

FIGURE 3.14

(a) An 8-bit gray-scale image of size 837 x 988 pixels. (b) through (i) Bit planes 8 through 1, respectively, where plane 1 contains the least significant bit. Each bit plane is a binary image. Figure (a) is an SEM image of a trophozoite that causes a disease called *giardiasis*. (Courtesy of Dr. Stan Erlandsen, U.S. Center for Disease Control and Prevention.)

17



Bit-Plane Slicing

FIGURE 3.15

Image reconstructed from bit planes: (a) 8 and 7; (b) 8, 7, and 6; (c) 8, 7, 6, and 5.

18

3.3 Histogram Processing

- Data-dependent pixel-based image enhancement method.
- Histogram = PDF of image pixels.
- Assumption: each image pixel is drawn from the same PDF independently (i.i.d.)
- Un normalized histogram

$$h(r_k) = n_k \quad \text{for } k = 0, 1, 2, \dots, L - 1$$

- Normalized histogram

$$p(r_k) = \frac{h(r_k)}{MN} = \frac{n_k}{MN}$$

19

Images and their histograms

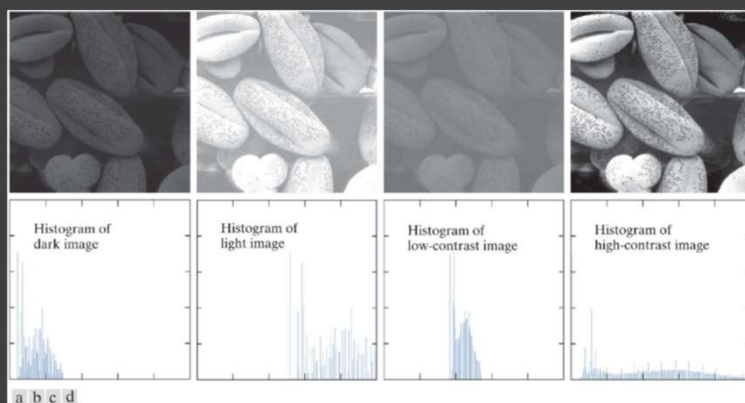


FIGURE 3.16

Four image types and their corresponding histograms. (a) dark; (b) light; (c) low contrast; (d) high contrast. The horizontal axis of the histograms are values of r_k and the vertical axis are values of $p(r_k)$

20

Why is Low Contrast a Problem?

- Important details can be hidden.
- Difficult for human eyes to interpret.
- Challenges for computer vision algorithms (e.g., feature detection, segmentation).
- Example: Medical images (X-rays, CT scans) where subtle details are critical for diagnosis.

21

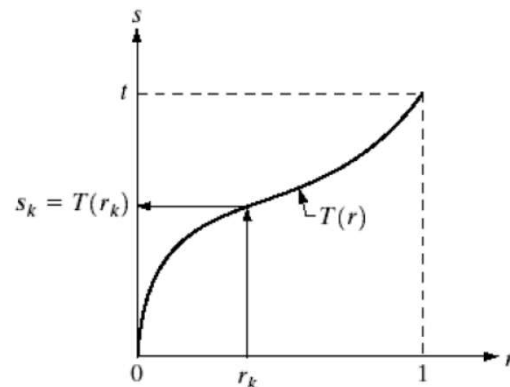
The Basic Solution: Histogram Equalization (HE)

- A technique to improve global contrast.
- Works by spreading out the most frequent intensity values.
- The goal is to make the histogram of the output image as uniform as possible.

22

Histogram Equalization

- A gray-level transformation method that forces the transformed gray level to spread over the entire intensity range.
 - Fully automatic,
 - Data dependent,
 - (usually) Contrast enhanced
- Usually, the discrete-valued histogram equalization algorithm does not yield exact uniform distribution of histogram.
- In practice, one may prefer "histogram specification"



$$s = T(r) = \int_{w=0}^r p_r(w) dw$$

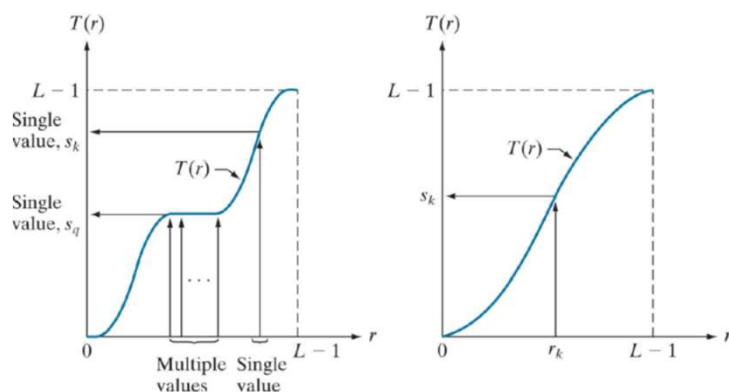
$$\Rightarrow p_s(s) = 1, \quad 0 \leq s \leq 1.$$

23

Histogram Equalization

FIGURE 3.17

- (a) Monotonic increasing function, showing how multiple values can map to a single value.
- (b) Strictly monotonic increasing function. This is a one-to-one mapping, both ways.



24

Histogram Equalization

An equalized histogram $\Rightarrow p_s(s) = 1, 0 \leq s \leq 1$.

$$\text{Equivalently, } F_S(s) = \begin{cases} 0 & s < 0; \\ s & 0 \leq s \leq 1; \\ 1 & s > 1. \end{cases}$$

In other words, $F_R(r) = F_S(s) = T(r) = s$

$$\text{Thus, } s = T(r) = F_R(r) = \int_{w=0}^r p_R(w)dw$$

25

Practical considerations

- $r, s \in \{0, 1, \dots, L-1\}$ instead of $[0, 1]$.
 $\Rightarrow$ integration is replaced by summation
- Assume # pixels in the image is N . # of pixels whose gray scale value is n_r . Then the mapping becomes

$$T(r) = \int_{w=0}^r p_R(w)dw$$

$$\Rightarrow s = \left\lceil \frac{L-1}{N} \sum_{w=0}^r n_w \right\rceil = \lceil (L-1) \cdot cdf(r) \rceil$$

the bracket indicates rounding to nearest integer.

26

Histogram Equalization

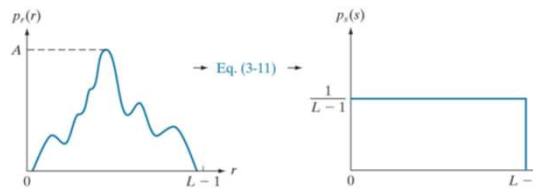


FIGURE 3.18

(a) An arbitrary PDF. (b) Result of applying **Eq. (3-11)** to the input PDF. The resulting PDF is always uniform, independently of the shape of the input.

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

27

Histogram Equalization Example

- Consider a 5×5 image with $L = 4$.

r	0	1	2	3
p(r)	6/25	7/25	7/25	5/25
Cdf(r)	6/25	13/25	20/25	25/25
s	1	2	2	3

0	0	1	1	2
1	2	3	0	1
3	3	2	2	0
2	3	1	0	0
1	1	3	2	2

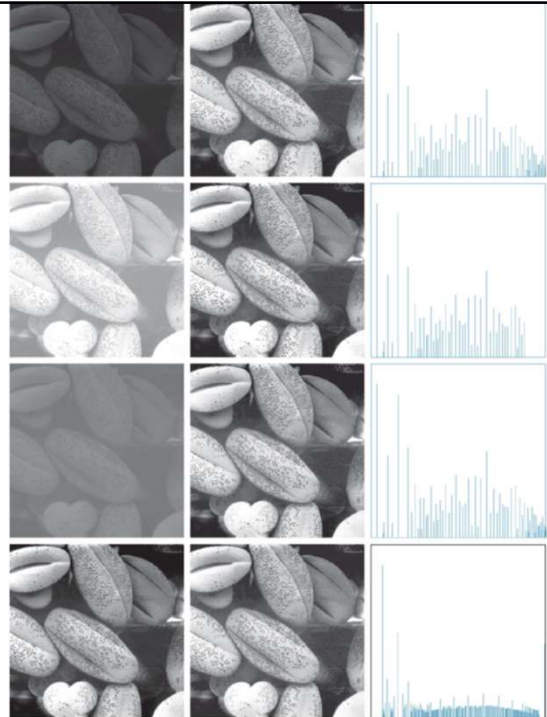
- Since original image already has an equalized histogram, the effect is not clear in this example.

1	1	2	2	2
2	2	3	1	2
3	3	2	2	1
2	3	2	1	0
2	2	3	2	2

28

Histogram equalization

FIGURE 3.20 Left column: Images from [Fig. 3.16](#) . Center column: Corresponding histogram-equalized images. Right column: histograms of the images in the center column (compare with the histograms in [Fig. 3.16](#)).



29

The Limitation with Global HE

- HE applies the same transformation to the entire image.
- This can over-enhance noise in uniform areas (e.g., sky, walls).
- It can also amplify contrast in already bright regions, leading to an unnatural or "blown-out" look.
- Example:
 - If a picture has a bright sky and a dark foreground, HE might make the sky look unnaturally bright to enhance the foreground details.

30

A Better Approach: Adaptive Histogram Equalization (AHE)

- AHE solves the global problem by working on small regions (tiles) of the image.
- The image is divided into a grid of non-overlapping tiles.
- HE is applied to each tile independently.
- To avoid artificial boundaries, a process called **bilinear interpolation** is used to blend the results between tiles.

31

The Problem with AHE

- While AHE solves the global issue, it can still over-amplify noise, especially in regions with little variation (e.g., smooth surfaces like a wall).
- This happens because the histogram of a uniform region is very narrow and tall, and equalizing it will spread out the small amount of noise, making it highly visible.

32

Introducing CLAHE: The Solution

- **Full Name:** Contrast Limited Adaptive Histogram Equalization
- CLAHE is an improvement on AHE.
- It addresses the noise amplification problem by introducing a "**contrast limit.**"
- Before equalizing the histogram of each tile, any histogram bin that exceeds a defined limit is "clipped."
- The clipped pixels are then distributed evenly across all the histogram bins.
- This prevents the over-enhancement of contrast and effectively limits noise amplification.

33

How CLAHE Works (Step-by-Step)

1. **Divide the Image:** Split the image into a grid of rectangular tiles.
2. **Calculate Histograms:** Compute the histogram for each tile.
3. **Clip Histograms:** Apply a clipping limit. Any bin exceeding this limit is clipped, and the excess pixels are redistributed.
4. **Equalize:** Equalize the clipped histogram for each tile.
5. **Interpolate:** Use bilinear interpolation to blend the equalized tiles and remove grid-like artifacts.

34

CLAHE vs. Other Methods (Comparison)

	Global HE	AHE	CLAHE
Pros	Simple, fast	Good for local contrast	Excellent for local contrast, limits noise amplification
Cons	Over-enhances noise, ignores local contrast	Can amplify noise	Computationally more intensive than HE

- CLAHE is the most effective method for improving **local image contrast** without amplifying noise.
- It's an adaptive technique that works on local regions.
- The key feature is the **contrast limit** that prevents over-enhancement.

35

Histogram Modification

One may want to convert the histogram to a target histogram that is not uniformly distributed. Rather with a new pdf $g(s)$.

In this case, one has $F_R(r) = F_S(s) = T(r) = g(s)$

$$\text{Thus, } T(r) = s = g^{-1}(F_R(r)) = g^{-1}\left(\int_{w=0}^r p_R(w)dw\right)$$

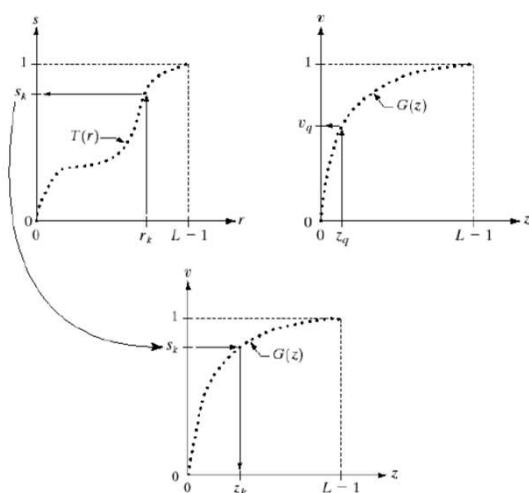
Assuming $g^{-1}(\cdot)$ exists over $[0 \ 1]$.

36

Histogram Matching

- Transform pdf of r to a desired pdf $p_s(s)$.
- A generalization of histogram equalization.
- Basic idea: Given $p_r(r)$ and desired pdf $p_z(z)$, find a transform $z = T(r)$, such that $P(Z \leq z) = P(R \leq r)$.

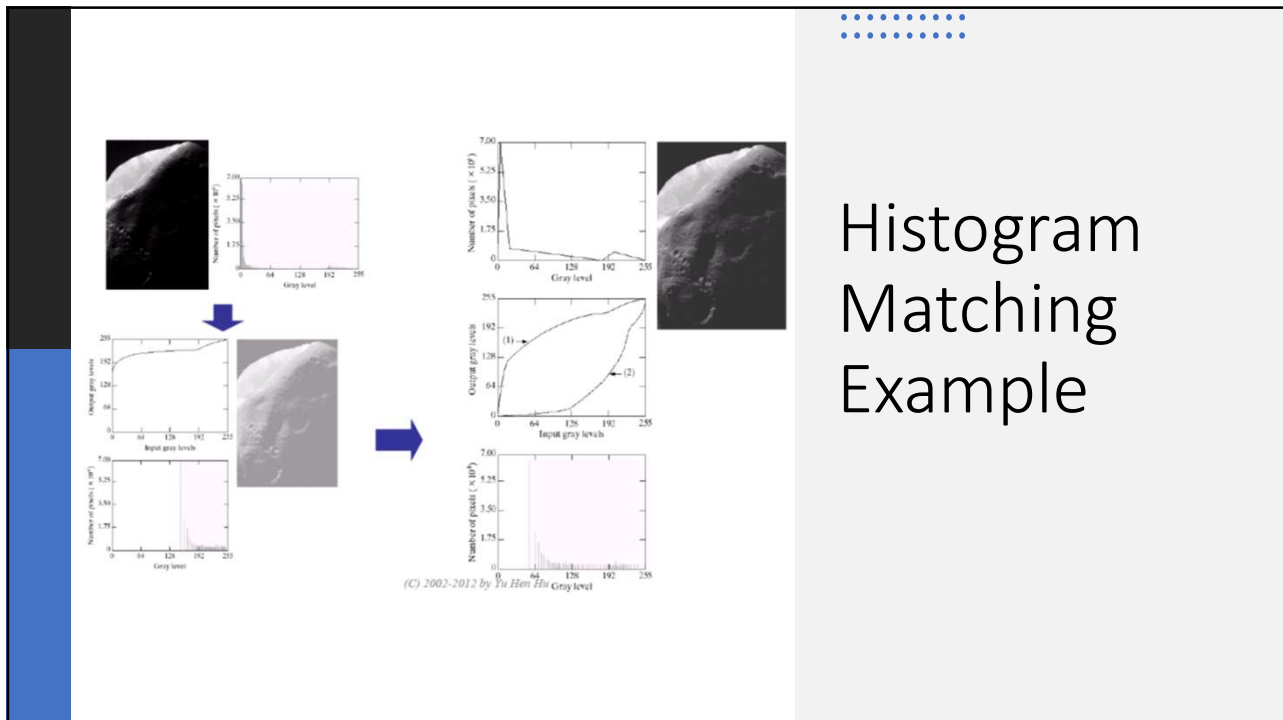
37



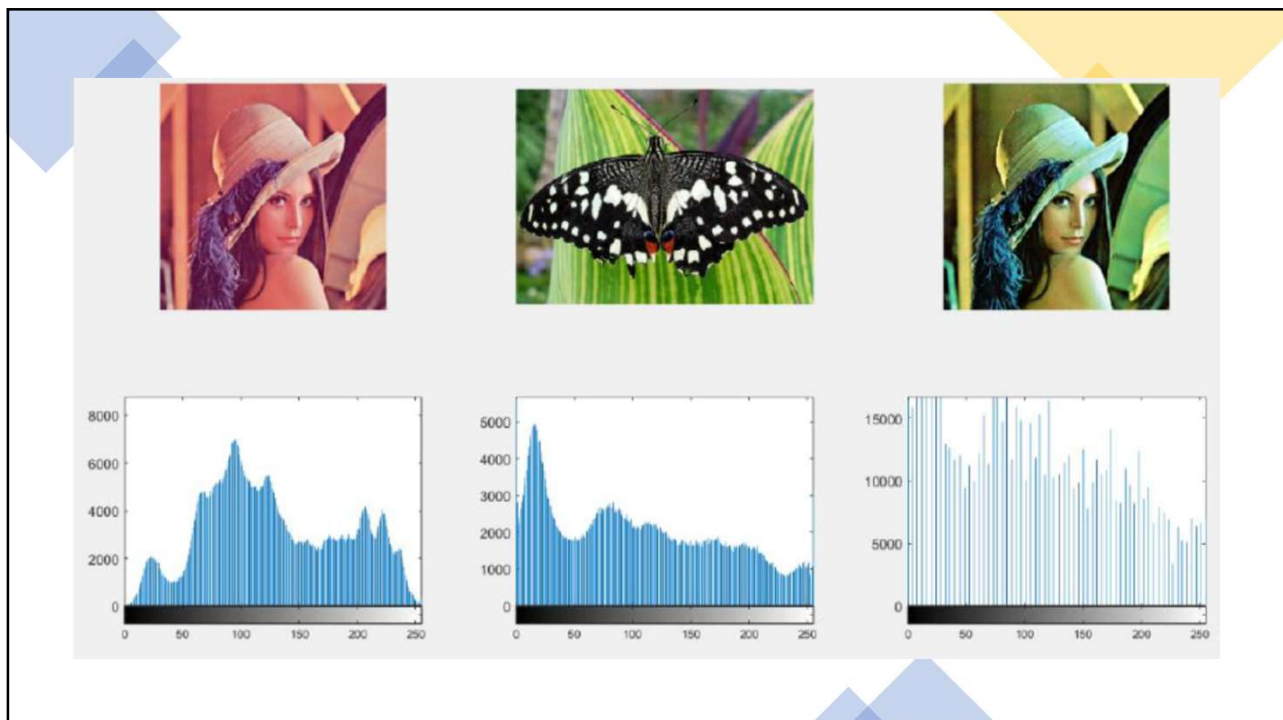
Indirect Method

- First equalize the histogram using transform $s = T(r)$.
- Equalize the desired histogram $v = G(z)$.
- Set $v = s$ to obtain the composite transform $z = G^{-1}(T(r))$

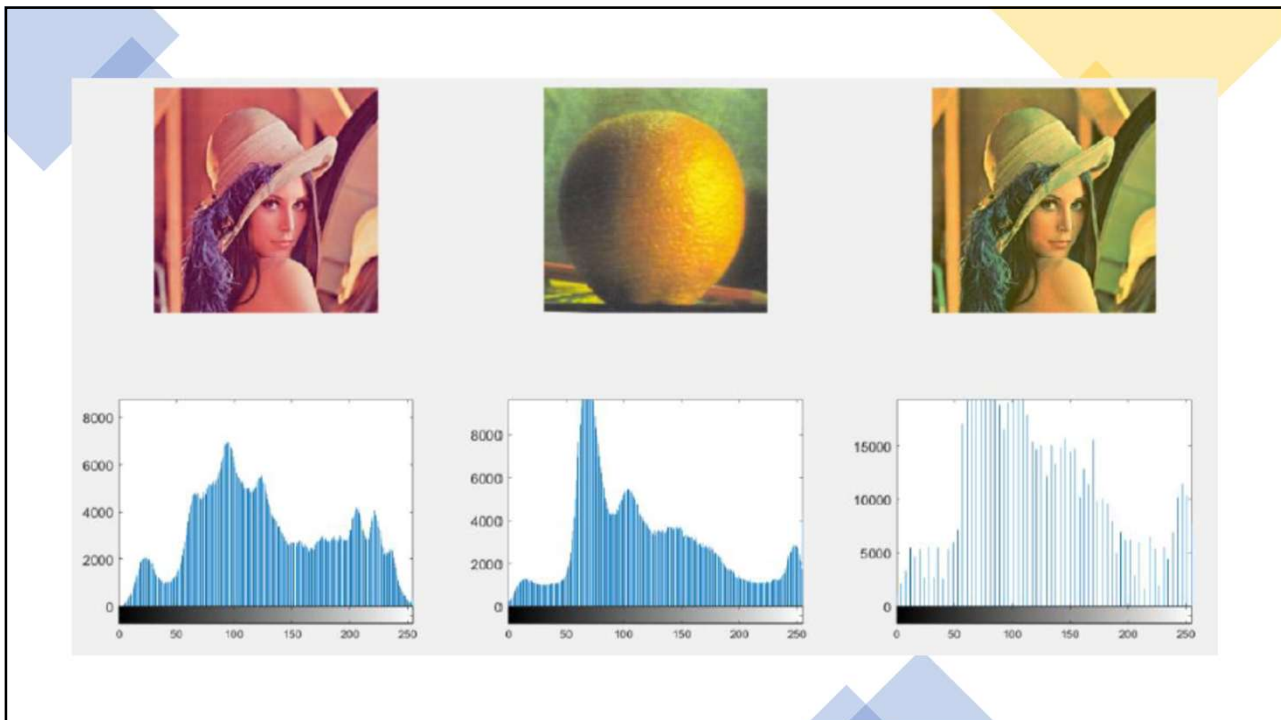
38



39



40

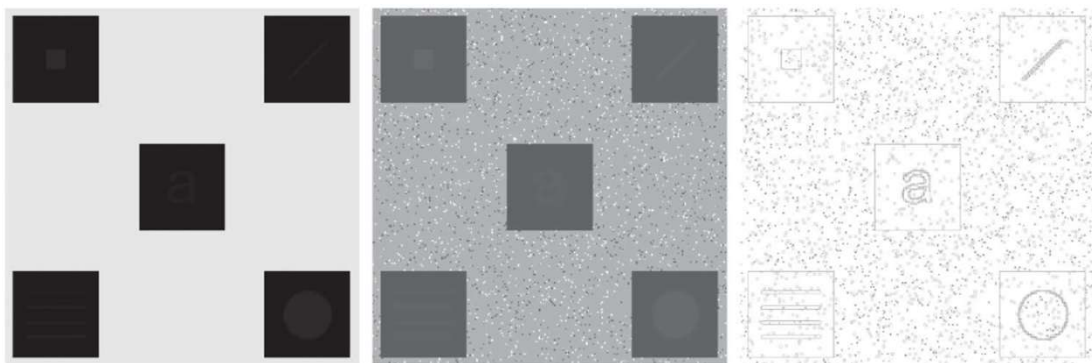


41

Local Histogram Processing

FIGURE 3.32

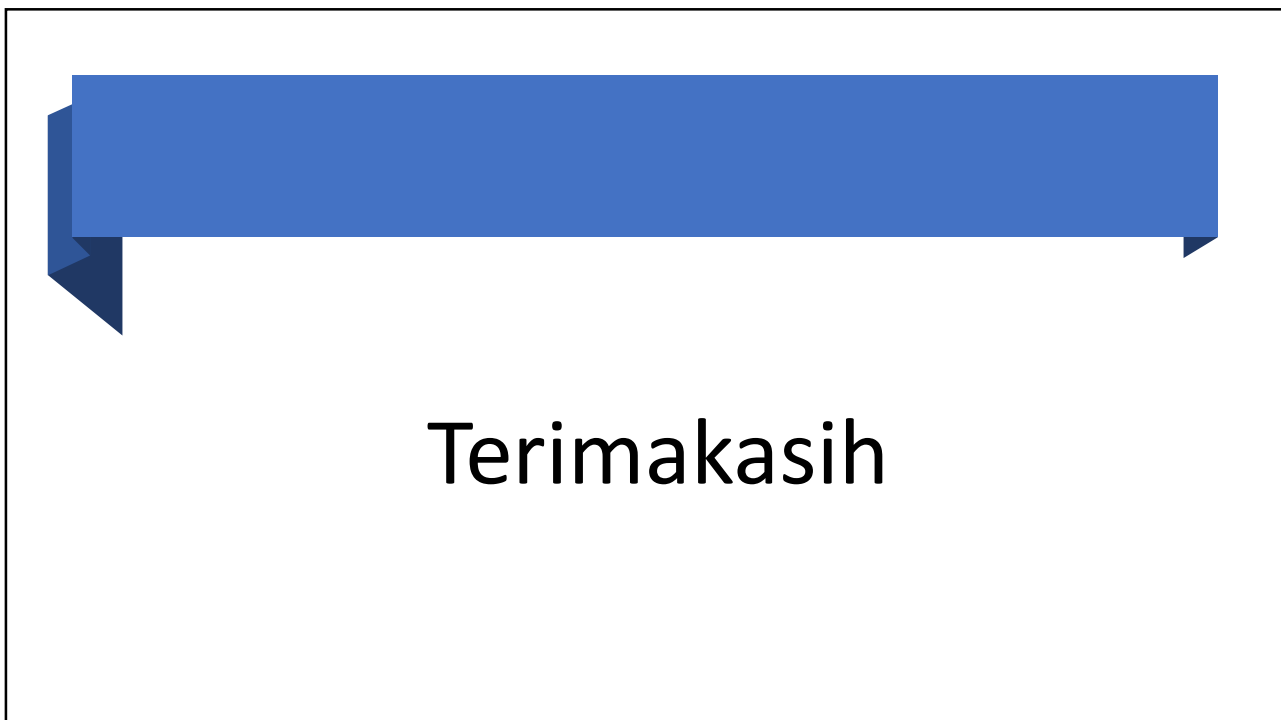
(a) Original image. (b) Result of global histogram equalization. (c) Result of local histogram equalization.



42



43



44